

Executive Summary

- **Cheapest option:** Oracle Cloud's Always Free tier (2 × A1.Arm instances, 2 OCPUs, 12 GB) effectively costs \$0 【70†L54-L63】. It offers generous free CPU/RAM but requires learning OCI and using ARM (or AMD micro) VMs.
- **Best value:** Hetzner Cloud CX11 (~€3.79/mo, 1 vCPU, 2 GB, 20 GB SSD, 20 TB) 【32†L42-L51】. Very low price for high bandwidth, but no India region. Contabo Cloud S is also cheap (~€4.50, 4 vCPU, 8 GB, 50 GB) 【51†L18-L27】.
- **Easiest to deploy:** AWS Lightsail (\$5/mo, 2 vCPU, 1 GB, 40 GB SSD) 【62†L898-L905】 – user-friendly UI, built-in static IP and snapshots. DigitalOcean (\$6/mo, 1 vCPU, 1 GB, 25 GB SSD) 【9†L146-L152】 and Linode (\$5/mo, 1 vCPU, 1 GB, 25 GB SSD) 【23†L49-L52】 are similarly easy.
- **India latency:** Vultr and DigitalOcean have Mumbai/Bangalore data centers; Linode (Akamai) has Mumbai. AWS (Mumbai), GCP (Mumbai), Oracle (Mumbai) are available. Hetzner/OVH/Scaleway lack India zones.

Provider (Plan)	vCPU / RAM	Storage (type)	Bandwidth	Price (USD/mo) / ₹	DC Regions (India)	IPv4	Docker & SSH	Backups
Hostinger (KVM1)	1 / 4 GB	50 GB NVMe SSD	4 TB	~\$6.49 (promo) / ~₹550 【5†L107-L115】	EU (GER)	Yes	Yes	Weekly
DigitalOcean (Basic 1 GB)	1 / 1 GB	25 GB SSD	1 TB	\$6.00 / ~₹500 【9†L146-L152】	SG, BLR (India)	Yes	Yes	Optional
Vultr (Regular 25 GB)	1 / 1 GB	25 GB SSD	1 TB	\$5.00 / ~₹425 【19†L173-L181】	SG, BLR, DEL	Yes	Yes	Snapshots
Linode (Akamai) (Nanode)	1 / 1 GB	25 GB SSD	1 TB	\$5.00 / ~₹425 【23†L49-L52】	Mumbai	Yes	Yes	Snapshots
Hetzner (CX11)	1 / 2 GB	20 GB SSD	20 TB	€3.79 (~\$4.4) / ~₹370 【32†L42-L51】	GER, FIN (EU)	Yes	Yes	Snapshots/ Backups

Provider (Plan)	vCPU / RAM	Storage (type)	Bandwidth	Price (USD/mo) / ₹	DC Regions (India)	IPv4	Docker & SSH	Backups
Scaleway (START1-XS)	1 / 1 GB	Flexible (25 GB NVMe)	100 Mbps	€5.00 (~\$5.5) / ~₹460 【44†L50-L54】	Paris, AMS	Yes	Yes	-
OVHcloud (VPS-1)	2 / 4 GB	40 GB NVMe	Unlimited	\$4.54 / ~₹390 【46†L41-L49】	UK, FR, USA (no IN)	Yes	Yes	Daily
Contabo (VPS S)	4 / 8 GB	50 GB NVMe	32 TB	€4.50 (\$5.3) / ~₹440 【51†L18-L27】	DE	Yes	Yes	Snapshots
UpCloud (1 GB Dev)	1 / 1 GB	10 GB SSD	Fair-use	\$3.50 / ~₹300 【55†L514-L518】	SG, HEL (no IN)	Yes	Yes	Optional
Lightsail (AWS)	2 / 1 GB	40 GB SSD	2 TB	\$5.00 / ~₹425 【62†L898-L905】	Mumbai, others	Yes	Yes	Snapshots
Google Cloud (Free)	~0.25 / 0.6 GB	30 GB HDD (free)	1 GB egress	Free tier 【79†L451-L458】	Mumbai, others	Yes	Yes	-
Oracle Cloud (Always Free)	2 / 12 GB*	2×47 GB Block (free)	-	Free tier 【70†L54-L63】 【70†L112-L119】	Mumbai, others	Yes	Yes	-

*> Always Free: 2 Arm A1 (total 2 OCPU, 12 GB) or 2 AMD micro (0.125 OCPU, 1 GB each) 【70†L54-L63】 【70†L112-L119】 .

Key Provider Details

- **Hostinger VPS (KVM):** Managed KVM-VPS, full root/SSH. Weekly automated backups included. No India DC; GDPR-compliant EU. 1 Gbps network. 99.9% SLA. Good docs; unmanaged Linux (you do OS security).
- **DigitalOcean:** True cloud droplet (KVM/QEMU), 24×7 support. 99.99% SLA. Control panel and CLI allow automated snapshot backups (20% of droplet cost). Free IPv4+IPv6. Data centers include Mumbai for low latency. Cloud Firewall, private networking available.

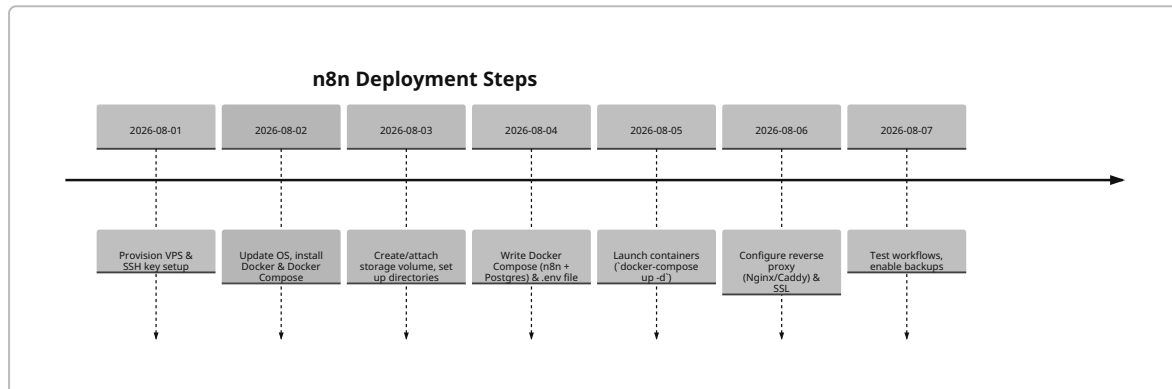
- **Vultr:** Offers “Regular” (shared SSD), “High-Frequency” (NVMe) plans. Cheapest (\$3.50) plan is 512 MB (IPv4) 【19†L122-L130】 ; \$5 plan (1 GB, 25 GB SSD) is recommended for n8n 【19†L173-L181】 . IPv4 and IPv6. Locations include Singapore, Mumbai, Bengaluru, New Delhi, etc. Unmanaged. 100% SLA (for premium plans). Block storage at \$1/GB.
- **Linode (Akamai Cloud):** Nanode plan (1 GB, 1 CPU, 25 GB SSD) 【23†L49-L52】 . 99.99% SLA, data centers in Mumbai, Singapore, Chennai, etc. Allows private IPs, VLAN. Snapshots available. Great docs and community. Full root access, unmanaged.
- **Hetzner Cloud:** CX (Cost-Optimized) series: CX11 ~ €3.79 (1 vCPU, 2 GB, 20 GB, 20 TB) 【32†L42-L51】 . Extremely cost-effective but only EU (Germany/Finland). High 1 Gbps link, 20 TB included traffic. IPv6 + IPv4. Flexible block storage and load balancers. Lacks native firewall, but can use Linux UFW. No direct support in India.
- **Scaleway:** START1 instances (1 vCPU, 1 GB, flexible NVMe) at €5 【44†L50-L54】 . Unlimited bandwidth (100 Mbps cap). EU-only (France/Netherlands). Instance needs flexible volume mount. Offers IPv4 and IPv6. No managed backups by default (you manually snapshot).
- **OVHcloud VPS (2027 series):** VPS-1 (2 vCPU, 4 GB, 40 GB NVMe) from \$4.54 【46†L41-L49】 . Daily snapshot backups included, anti-DDoS built-in, 500 Mbps-1 Gbps network. Unlimited traffic. Data centers in EU, US; no India. 99.95% SLA.
- **Contabo:** Core VPS S (~€4.50, 4 vCPU, 8 GB, 50 GB NVMe) 【51†L18-L27】 . Unmanaged, EU. 600 Mbps shared bandwidth, 30 TB (or more) included. Optional weekly snapshots. Known for massive RAM for price.
- **UpCloud:** Premium performance. Old dev tier 1 GB at \$3.50 【55†L514-L518】 (limited to 2 servers, likely phased out). Current Cloud Native min is 4 GB for \$14 【55†L422-L424】 . 99.99% SLA, multiple global DCs (Singapore, etc). High IOPS MaxIOPS storage (\$0.25/GB). Offers firewall, floating IPs, private networks. Pricier but very reliable. No India region.
- **AWS Lightsail:** Simplified VPS. \$5/mo includes static IP, auto DNS, up to 7 static IPs. 1 GB, 2 vCPU, 40 GB SSD, 2 TB transfer 【62†L898-L905】 . Easy console and CLI. 99.99% SLA, multi-region (Mumbai available). Integrated backup snapshots. Built-in firewall, AWS security groups. Good for quick deploys (Ubuntu image with one-click containers).
- **Google Cloud (Compute Engine):** Always Free: 1 × e2-micro (shared 2 vCPU burst, ~0.6 GB) + 30 GB HDD + 1 GB egress 【79†L451-L458】 . Only US free regions. For production, smallest paid e2-small (2 vCPU, 2 GB). 99.95% SLA. Multi-Asia DCs (Mumbai). Enduring use discounts. Includes IPv4+IPv6.
- **Oracle Cloud:** Always Free: two Ampere A1 Flex (2 OCPUs total, 12 GB RAM) or two AMD micros (each 1 GB) 【70†L54-L63】 【70†L112-L119】 , plus 2×47 GB block storage (200 GB always free). High-performance ARM CPUs (lower price). Data centers include Mumbai. 99.95% SLA. Note idle free VMs may be reclaimed if unused. Free load balancer option, free egress (tennis-ball network).

Recommended 3 Options

1. **Best Cheapest:** *Oracle Cloud Always Free*. Provides up to 2 ARM OCPUs/12GB (or 2×1GB AMD) at **\$0/mo** 【70†L54-L63】 【70†L112-L119】 . Free 2×47 GB boot disks and load balancer. Use OCI CLI or console to provision. Supports Docker. Indian region available (Mumbai). **Deployment note:** Use the A1 instance (Arm) for more RAM. Remember to keep instance active (idle VMs may be recycled).
2. **Best Value:** *Hetzner CX11 (Cost-Optimized)*. ~€3.79/mo (≈₹370) 【32†L42-L51】 for 1 vCPU, 2 GB RAM, 20 GB SSD, 20 TB BW. Incredible bandwidth/value in EU. Good for small/medium n8n. (If EU location is acceptable; otherwise Contabo is an alternative.) Comes with 20 TB free traffic. Unmanaged (full SSH) – suitable if comfortable on Linux.

3. **Easiest to Deploy:** *AWS Lightsail \$5 plan*. Simple UI, quick SSL (via Amazon Certificate Manager or Let's Encrypt), static IP and snapshots included. 2 vCPU, 1 GB, 40 GB [\[62†L898-L905\]](#) . Supports Ubuntu with Docker. Good AWS tutorials. Also consider DigitalOcean's \$6 droplet or Linode \$5 – very similar ease.

Deployment Steps (Mermaid Timeline)



1. **SSH & OS Prep:** Secure your instance (disable root login, enable firewall). Update packages (`apt update && apt upgrade`).
2. **Install Docker:** On Ubuntu/Debian: `apt install docker.io docker-compose` . Or use Docker's official script.
3. **Persistent Storage:** Attach any extra disk or use cloud volume for `/home/node/.n8n` (n8n's data) and for database if using local Postgres. Ensure volumes persist across reboots.
4. **Database:** Decide on DB:
5. **SQLite:** Default but **not recommended** beyond tiny projects [\[73†L159-L166\]](#) .
6. **Postgres:** Install via Docker (official `postgres` image) or use a managed service (like Supabase free Postgres). Set env vars `DB_TYPE=postgresdb` , `DB_POSTGRESDB_HOST` , etc. Remote DB (Supabase) offloads load and persists independently (good practice).
7. **Docker Compose:** Create `docker-compose.yml` with n8n container (exposing 5678) and reverse-proxy (Caddy or Nginx) for HTTPS. Example templates are in n8n docs. Mount the volumes and set environment (including `N8N_ENCRYPTION_KEY` , webhook URL, basic auth, timezone). [\[73†L123-L133\]](#) [\[73†L141-L146\]](#)
8. **Reverse Proxy & SSL:** Use a TLS-ready proxy. For example, Caddy can auto-provision Let's Encrypt certs. Or use Nginx + Certbot. Proxy domain (e.g. `n8n.example.com`) to the internal port (5678). Ensure firewall allows 80/443.
9. **Start n8n:** Run `docker-compose up -d` . Check logs (`docker logs n8n`) for errors.
10. **Basic Auth: Mandatory** for public n8n. Enabled via env `N8N_BASIC_AUTH_ACTIVE=true` .
11. **Backups:** Schedule periodic database dumps and file backups. e.g. daily `pg_dump` or volume snapshots (use provider's snapshot or `rsync` to object storage). Offsite backups (e.g. remote bucket) for disaster recovery.

n8n Database: SQLite vs PostgreSQL

- **SQLite:** Built-in, ok for tiny workloads ($\leq 5k$ executions/day, ≤ 5 GB DB) [\[73†L159-L166\]](#) . Struggles with concurrency and heavy load ("compact car"). Only use for testing or trivial use.

- **PostgreSQL (Self-Host or Supabase):** Recommended for production. Handles growth and concurrency (“SUV” in n8n terms [【73†L159-L166】](#)). If you use Supabase (hosted Postgres), n8n remains light and you get managed DB features. Supabase free tier (10 GB) is ample for many n8n setups.

Security & Scaling Tips

- **Firewall:** Restrict ports (open only 22, 80, 443, 5678 as needed). Use provider firewalls (DO Cloud Firewall, AWS SGs, etc) or Linux UFW.
- **Updates:** Regularly update OS and Docker images.
- **n8n Auth:** Always enable basic auth or OAuth. Do not expose n8n without auth.
- **Scaling:** For higher load, consider: multiple n8n instances with an external queue (Redis) or Kubernetes. Or use n8n’s [enterprise scaling guide](#).
- **Monitoring:** Use logs/healthchecks. Use a cron to restart on crash if not using `restart: unless-stopped`.
- **Cost-saving:** Use reserved/prepaid plans (annual) or spot instances for non-critical workflows. Choose lower-cost regions (e.g. OVH/Hetzner EU vs AWS US). Take advantage of free credits or free tiers (Oracle, GCP, AWS free trial).

Summary of Specs and Costs

- *Small workload:* ~1 vCPU, 1–2 GB RAM (monthly ~₹300–600). SQLite OK for demo; use Postgres for more reliability.
 - *Medium workload:* 2 vCPU, 4+ GB RAM (₹700–1200/mo). Always use PostgreSQL. Consider external database (Supabase) to reduce disk I/O and for daily backups.
 - *Monthly INR cost:*
 - Hostinger VPS-1: ~₹550/mo (promo) [【5†L107-L115】](#) .
 - Hetzner CX11: ~₹370/mo [【32†L42-L51】](#) .
 - Lightsail 1GB: ~₹425/mo [【62†L898-L905】](#) .
 - Contabo 8GB: ~₹440/mo [【51†L18-L27】](#) .
 - Oracle Free: ₹0 (with caution).
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